Threat Intelligence Aggregator (Non-AI)

Cybersecurity Internship Project Developed Using Python

Your Name | Organization Name

Made with GAMMA

Project Overview

Threat intelligence is critical for proactive defense against evolving cyber threats. Organizations receive intelligence from multiple sources, but the volume and inconsistency make manual analysis impractical.

This automated aggregator transforms raw threat feeds into actionable security intelligence, enabling faster response and better threat prioritization.

Collect Feeds

Gathers threat intelligence from multiple sources

Extract IOCs

Identifies indicators of compromise

Normalize Data

Standardizes formats for analysis

Correlate Threats

Links related indicators

Generate Blocklists

Creates defensive security controls

Produce Reports

Delivers actionable intelligence

Made with GAMMA

Problem Statement

Modern security operations face significant challenges in managing threat intelligence at scale:

Multiple Feed Sources

Security teams receive intelligence from dozens of vendors, each with unique formats and update schedules

Inconsistent Formats

Feeds arrive in various formats including TXT, CSV, JSON, and XML, requiring different parsing logic

Data Duplication

Same indicators appear across multiple feeds, creating noise and confusion in threat assessment

Manual Analysis Bottleneck

Human analysts cannot process thousands of indicators quickly enough to prevent active attacks

Security teams need automated solutions to normalize, correlate, and prioritize threat intelligence for effective defensive action.

Made with GAMMA

Project Objectives

01

Load IOC Feeds

Read threat intelligence from file sources

02

Parse Indicators

Extract IPs, domains, URLs, and cryptographic hashes

03

Normalize Data

Convert all indicators to unified format

04

Correlation Engine

Detect repeated indicators across sources

05

Assign Severity

Mark threats as Low or High risk

06

Generate Blocklists

Create firewall, web filter, and endpoint rules

07

Export Report

Produce comprehensive threat intelligence summary

Made with GAMMA

What Are Indicators of Compromise (IOCs)?

IOCs are forensic evidence that reveals malicious activity in your network. They help security teams detect and respond to threats quickly.

Malicious IP Addresses

IP addresses associated with attackers, command-and-control servers, or scanning activities

Example: 192.168.184.132

Suspicious Domains

Domain names used for phishing, malware distribution, or malicious infrastructure

Example: malware-download[.]xyz

Dangerous URLs

Complete web addresses that host malware, exploit kits, or credential harvesting pages

Example: evil-site[.]com/exploit.exe

Malware File Hashes

Cryptographic fingerprints (MD5, SHA-1, SHA-256) identifying malicious executable files

Example: e3b0c44298fc1c149afbf4c8996fb924

Made with GAMMA

System Architecture & Workflow

Load Feeds

Read threat intelligence files from multiple sources

Parse Indicators

Extract IPs, domains, URLs, and hashes using regex patterns

Normalize Data

Standardize formats, validate syntax, remove duplicates

Correlation Engine

Detect indicators appearing in multiple feeds, assign risk scores

Generate Blocklists

Create firewall rules, web filter lists, endpoint protection signatures

Export Final Report

Generate summary with statistics and actionable intelligence

Made with GAMMA

Correlation Engine: The Core Logic

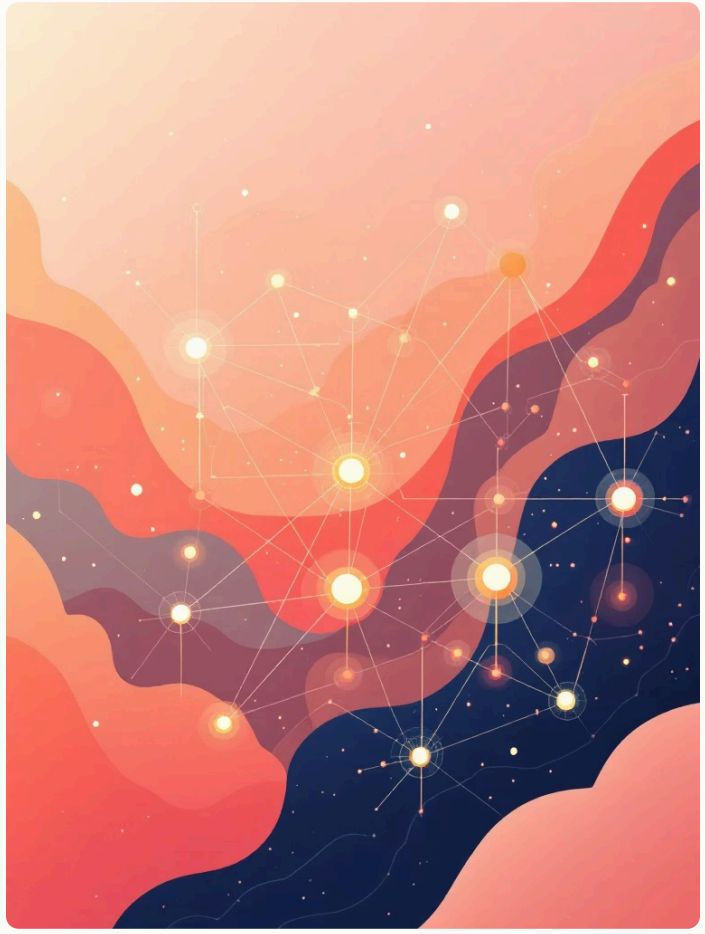
The correlation engine is the intelligence behind this system. It analyzes which indicators appear across multiple threat feeds, using frequency as a proxy for threat severity.

How It Works

- Tracks each indicator's appearance count across all feed sources
- Indicators appearing in 2+ feeds marked as **High Risk**
- Single-feed indicators marked as **Low Risk**
- Generates prioritized threat list for security teams

Why Repeated Indicators Are More Dangerous

When the same IP, domain, or hash appears in multiple independent threat feeds, it indicates widespread malicious activity. These indicators represent confirmed threats used by multiple attacker groups, making them higher priority for blocking and investigation.



Made with GAMMA

Blocklist Generation for Defensive Security

Blocklists are the actionable output that transforms intelligence into prevention. These files can be directly imported into security controls.

IP Blocklist

File: blocklist_ips.txt

Format: One IP per line (CIDR notation supported)

Usage: Firewall rules, network ACLs, intrusion prevention systems

Example: 192.168.184.132

Domain/URL Blocklist

File: blocklist_domains.txt

Format: One domain or URL per line

Usage: Web proxies, DNS filtering, content filters

Example: malware-download[.]xyz

Hash Blocklist

File: blocklist_hashes.txt

Format: One hash per line (MD5/SHA-1/SHA-256)

Usage: Endpoint protection, file integrity monitoring, EDR solutions

Example: e3b0c44298fc1c149afbf4c8996fb924

Security Impact: Blocklists create preventive controls that stop known threats before they can execute, reducing incident response workload and improving security posture.

Made with GAMMA

Example Output

Console Summary

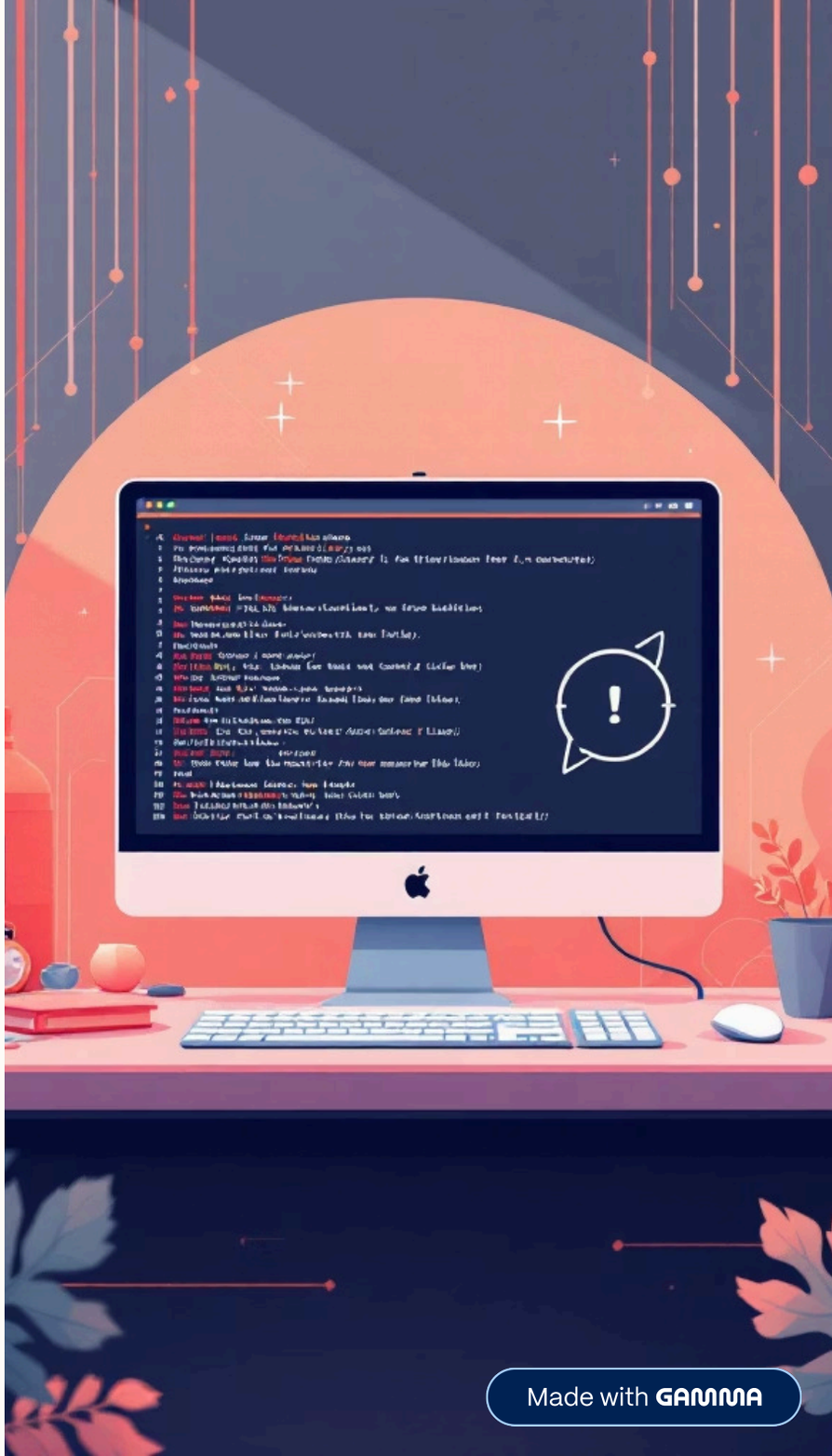
```
Processing threat feeds...
Found 1428 indicators across 5 feeds
Normalizing data...
Correlating threats...
Generating blocklists...
Writing final report...

SUMMARY:
Total indicators processed: 1428
High-risk repeated indicators: 87
Generated blocklists: 3
Report saved: threat_report_2024.txt
```

Generated Files

- `blocklist_ips.txt` - 423 malicious IP addresses
- `blocklist_domains.txt` - 345 suspicious domains
- `blocklist_hashes.txt` - 210 malware file hashes
- `threat_report_2024.txt` - Full analysis with statistics

Report Highlights

- 87 indicators appeared in 2+ feeds (High Risk)
- 5 unique feed sources processed
- 28% reduction in duplicate entries



Made with GAMMA

Conclusion & Learning Outcomes

Built Automated Threat Intelligence Aggregator

Created a production-ready system that processes multiple threat feeds without human intervention

Implemented IOC Parsing & Normalization

Developed robust parsing logic for diverse feed formats and standardized data structures

Created Correlation Engine

Designed intelligent risk-scoring algorithm that prioritizes threats based on appearance frequency

Generated Actionable Blocklists

Produced ready-to-deploy security controls for firewalls, web filters, and endpoint protection

Learned SOC-Level Defensive Workflow

Understood the complete threat intelligence lifecycle from collection to implementation

> **This project demonstrates practical application of cybersecurity principles, preparing interns for real-world security operations center responsibilities and threat intelligence analysis.**

Made with GAMMA